

JIAYI WU

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🔍 RESEARCH INTERESTS

Federated Learning, Privacy-preserving Machine Learning, AI Security, Applied Cryptography, Trustworthy AI

🎓 EDUCATION

Xidian University, Xi'an, China Sep. 2023 – Jun. 2026

- M.Eng. in Electronic Information, School of Telecommunications Engineering. Affiliated with the **State Key Laboratory of Integrated Services Networks**. GPA: 87.67/100. Honors: Special Scholarship of Xidian University.
- Core Courses: Modern Public-key Cryptography and Application, Engineering Optimization Methods.

Northwest University, Xi'an, China Sep. 2019 – Jun. 2023

- B.Eng. in Electronic Information Science and Technology. GPA: 86.6/100. Honors: Outstanding Graduate of Shaanxi Province, Outstanding Graduate of Northwest University, First-Class Scholarship (3 times).
- Core Courses: Digital Electronic System, Signals and Systems

University of Essex, Colchester, United Kingdom Sep. 2019 – Jun. 2023

- B.Sc. in Electronic Systems Engineering (**First Class Honours**), joint program with Northwest University. GPA: 76/100.
- Core Courses: Network Fundamentals, Computer Security, Advanced Embedded Systems Design.

💼 PROFESSIONAL EXPERIENCE

Tencent Technology Co.Ltd | *Backend Engineer* Jul. 2026 – Present

- Develop backend services for WeData Data Map, an enterprise-scale metadata management and data catalog platform supporting data discovery, governance, and analytics workflows.
- Implement backend features and RESTful APIs using Java, Spring Boot, MySQL, and Redis to support metadata management and data service functionalities.
- Integrate LLM-assisted programming workflows into daily development for rapid prototyping, code review, debugging, and technical documentation.

Huawei Technologies Co., Ltd. | *Intern, Network Security and Privacy Protection* Jul. 2025 – Sep. 2025

- Worked on **API anomaly detection** for network security, combining feature engineering and machine learning methods to identify abnormal API access, traffic bursts, and malicious calling behaviors.
- Built **user-level**, **traffic-level**, and **behavior-level** detection models using Autoencoders, Isolation Forest, Z-score analysis, Markov Chains, and LSTM. Extracted features such as URL thresholds, access frequency, and error rates, achieving a **23% reduction** in false positives.

📖 PUBLICATIONS

* denotes first author

Manuscripts Under Review

1. **Jiayi Wu***, et al. "FedCuckoo: Cuckoo Hashing Based Privacy-Preserving Personalized Federated Learning with Dynamic Parameter Selection." Under review at *IEEE Transactions on Mobile Computing*. (SCI Q1, CCF-A)

Selected Publications

2. **Jiayi Wu***, et al. "Strawberry Disease Detection Through an Advanced Squeeze-and-Excitation Deep Learning Model." *IEEE Transactions on AgriFood Electronics*, vol. 2, no. 2, pp. 259–267, 2024.
3. **Jiayi Wu***, et al. "Plant Disease Detection: Electronic System Design Empowered with Artificial Intelligence." In *Proceedings of the IEEE Conference on AgriFood Electronics*, Torino, Italy, pp. 30–34, 2023.
4. **Jiayi Wu***, et al. "Smart Guide Trolley Mango: Smart Blind Guide Device Based on Raspberry Pi." In *Proceedings of the IEEE International Conference on Electrical Engineering, Big Data and Algorithms*, Changchun, China, 2022.

</> PROJECT EXPERIENCE

Privacy Preservation in Personalized Federated Learning

Sep. 2023 – Jun.2026

- Conducted research on privacy-preserving personalized federated learning under non-IID settings, focusing on dynamic personalization and secure collaborative training.
- Proposed **FedCuckoo**, which combines dynamic parameter selection with Cuckoo hashing to protect the positions of personalized parameters during transmission.
- Further developed **FedFlock**, which supports grouped aggregation of personalized parameters among similar clients and integrates anonymous identities with Paillier encryption for privacy protection.
- Implemented the proposed methods in Python/PyTorch and evaluated them on benchmark datasets; this work has led to one first-author manuscript under review at *IEEE TMC*, with a second manuscript in preparation.

Secure and Trustworthy Data Circulation and Transaction

Jan. 2025 – Jun.2026

- Participated in a research project on secure and trustworthy data circulation and transaction, focusing on privacy-preserving computing and security mechanisms for multi-source heterogeneous data.
- Contributed to the design of trusted security privacy computing technology for multi-source heterogeneous data and privacy protection schemes for multi-source data circulation transactions.

Blockchain-based Commodity Supply Chain Financial Services

Sep. 2024 – Jun.2026

- Contributed to a provincial key R&D project on secure data aggregation and fair resource allocation in blockchain-enabled supply chain finance.
- Participated in the design of a dynamic contribution evaluation method for multi-party collaborative learning to support fair and secure cooperation among participants.

Plant Disease Detection Based on Computer Vision

Oct. 2022 – Apr. 2023

- Conducted research on deep learning-based plant disease detection, including model design, training, and performance evaluation for image classification tasks.
- This project is funded by **Innovate UK Knowledge Transfer Partnerships (KTPs)**. It is a collaborative project between the University of Essex and Wilkin & Sons Ltd.;
- Improved disease recognition performance by enhancing a ResNet-based model with squeeze-and-excitation mechanisms; the resulting work led to two first-author publications.

ACADEMIC SERVICE & TEACHING

- **Teaching Assistant**, Xidian University, 2024 – Present.
- **Reviewer**, IEEE VTC 2025-Spring.

SELECTED LEADERSHIP

Student Union, School of Information Science and Technology | *Chairman*

Sep. 2019 – Jun. 2022

- Coordinated large-scale campus activities, and promoted organizational planning and communication across teams.

Class 1, Electronic Information Science and Technology | *Class Monitor*

Sep. 2019 – Jun. 2023

- Organized class affairs and academic activities, supporting class management, and the development of a strong learning environment.

HONORS & AWARDS

- National Bronze Award, China International College Students' Innovation and Entrepreneurship Competition
- Third Prize, Northwest Regional Division, China College Students Computer Design Competition
- Second Prize, Shaanxi Division, National College Students Mathematical Modeling Competition

SKILLS

- **Programming & Tools:** Python, Java, C/C++, SQL, Redis, PyTorch, Git, LaTeX
- **AI-assisted Development:** Claude Code, CodeBuddy, LLM-assisted debugging and code generation
- **Languages:** English (IELTS 6.5, CET-6), Chinese (native)